Ben-Gurion University of the Negev

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Fourth Year Engineering Project

PDR
Project name – AskUni : A Smart Assistant for Student Life

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1 Abstract

In recent years, the university's digital ecosystem has become increasingly fragmented, encompassing multiple platforms such as faculty websites, course registration systems, personal portals, and administrative interfaces. This fragmentation has created an inconsistently structured information environment, where effective navigation often requires prior institutional knowledge and considerable time investment. The challenge is particularly pronounced for new users - such as incoming students and prospective applicants - who frequently struggle to locate essential forms, procedures, and academic resources. Although the university website currently offers a basic keyword-based search function, its performance and accuracy remain limited, often yielding incomplete or poorly structured results.

This shortcoming underscores the need for a more intelligent and context - aware navigation framework. To address these challenges, this project proposes the development of an AI - driven intelligent navigation and search system that enables seamless interaction with the university's digital resources. Leveraging LLMs and Vector Search technologies, the system will analyze and index web content across university platforms, interpret user queries in natural language, and provide precise, contextually relevant responses. The goal is to enhance user experience by reducing search time, improving information accessibility, and enabling more efficient exploration of the university's online environment.
Keywords: LLM, Vector Search, Smart Navigation.

1.1 תקציר

בשנים האחרונות הפכה סביבת המידע הדיגיטלית של האוניברסיטה למבוזרת ומורכבת יותר, וכוללת מגוון רחב של פלטפורמות כגון אתרי פקולטות, מערכות רישום לקורסים, פורטלים אישיים וממשקים מנהליים. פיצול זה יצר סביבה שבה מבנה המידע אינו עקבי, והניווט בין המערכות דורש ידע מוקדם על מבנה האוניברסיטה והשקעת זמן משמעותית. הקושי בולט במיוחד בקרב משתמשים חדשים- כגון סטודנטים חדשים ומועמדים ללימודים אשר מתקשים באיתור טפסים, נהלים ומשאבים אקדמיים חיוניים.

למרות שהאתר האוניברסיטאי כולל כיום מנוע חיפוש בסיסי המבוסס על מילות מפתח, ביצועיו מוגבלים והוא מספק לעיתים תוצאות חלקיות או לא ממוקדות. מגבלה זו מדגישה את הצורך במסגרת ניווט חכמה ומבוססת הקשר. כדי להתמודד עם אתגרים אלו, מוצע הפרויקט הזה לפתח מערכת חכמה לניווט וחיפוש מבוססת בינה מלאכותית ,שתאפשר אינטראקציה טבעית עם מקורות המידע הדיגיטליים של האוניברסיטה.

באמצעות שילוב בין (LLM) ו- Vector Search המערכת תבצע ניתוח ואינדוקס של תכנים מאתרי האוניברסיטה, תפרש שאילתות בשפה טבעית ותספק תשובות מדויקות ורלוונטיות להקשר.

מטרת המערכת היא לשפר את חוויית המשתמש, לצמצם את זמני החיפוש, להגדיל את הנגישות למידע, ולאפשר ניווט יעיל ואינטואיטיבי יותר במערכות המקוונות של האוניברסיטה.

מילות מפתח: LLM, חיפוש וקטורי, ניווט חכם.

2 The purpose of this project

This tool can be applied to several use cases. Its primary application is intelligent navigation across the university's fragmented web platforms, providing a seamless and unified way to access information distributed among different university systems.
The project aims to create a unified, AI-driven interface that enables users to interact with the university's online systems through natural language queries.

By integrating Large Language Models (LLMs) with Vector Search technology, the system will be able to understand user intent and provide relevant, accurate, and concise answers drawn from multiple sources within the university's digital ecosystem.
Ultimately, the project seeks to enhance the overall user experience, reduce search time, and improve accessibility and efficiency when navigating university resources.

Success Metrics ,to evaluate the system's performance, the following Key Performance Indicators have been defined. (For detailed testing methodologies and acceptance criteria, please refer to Section 8).
Accuracy: Achieve at least 80% accuracy on domain-specific queries.
Latency: Maintain a response time of under 3.5 seconds.
Concurrency: Support multiple simultaneous users without service degradation.
User Satisfaction: Achieve a positive rating of over 80% in efficiency and usability surveys.

3 Project specification sheet

AskUni is a student-facing assistant that answers university-related questions in natural language and provides direct links to authoritative university sources, with the goal of reducing the time and confusion caused by fragmented information spread across multiple university sites and sub-sites. the system is built by collecting official university web content (public pages and/or explicitly permitted sources), extracting and cleaning the text, segmenting it into retrievable chunks, and indexing those chunks semantically using embeddings and a vector similarity index (a vector database or equivalent). when a user submits a query, AskUni retrieves the most relevant passages first and then generates a concise response grounded in those passages (retrieval-augmented generation), while keeping metadata for traceability such as the page url, title, section heading, and last-updated time when available. the work is carried out in phases: defining requirements and mapping common user intents, implementing the ingestion pipeline, building the embedding and retrieval layer, implementing grounded response generation with link output and uncertainty

behavior, evaluating performance on representative questions, and iterating while maintaining a refresh strategy for content updates. typical applications include student services guidance (procedures, deadlines, forms, contacts), admissions and enrollment guidance, academic workflow support (registration steps, prerequisites, departmental rules), and reducing repetitive support workload by directing users to the right authoritative page. performance is specified in measurable terms: the input is a natural-language question and the output is a concise answer plus one or more supporting urls from authoritative sources; if sufficient evidence is not found, the system explicitly states uncertainty and returns the closest relevant authoritative links rather than fabricating procedures or details. quality is evaluated by grounding correctness (at least one returned url supports the main claim), answer accuracy against official sources on a defined test set, retrieval quality (top-k results contain clearly relevant passages), robustness to paraphrase (similar intents phrased differently lead to consistent sources and answers), and interactive latency appropriate for real use. operating constraints include indexing only permitted content and periodically refreshing the index to remain aligned with changing policies and pages; conceptually, the system flow is user query to query processing to vector retrieval to grounded answer generation to a response containing the answer and urls.

4 Development, and Work Phases

This section outlines the main phases of the system's development, from data collection and processing to deployment and evaluation.

4.1 Text Collection and Requirements Analysis

Collect relevant content from official university sources and analyze user needs to define main system requirements.

4.2 Data Preprocessing and Embedding Generation

Clean and segment text data into semantic units, remove irrelevant content, and generate embeddings representing each segment's meaning.

4.3 Vector Database Construction and Semantic Search

Store embeddings in a vector database (e.g., FAISS, Pinecone) and implement semantic search for meaning-based retrieval with optimized response time.

4.4 LLM Integration and Automated Answer Generation

Integrate an LLM (e.g., GPT) with the vector search to produce coherent, context-aware responses and deploy a functional AI chatbot on the university website.

4.5 User Interface Development and System Testing

Develop an intuitive web interface, conduct user testing, and refine the system for improved

accuracy and usability.

4.6 Results Evaluation and Feature Expansion

System performance will be evaluated based on answer accuracy, response time, and overall efficiency compared to traditional keyword-based search. Following evaluation, advanced navigation features such as direct links and step-by-step guidance will be added, and system coverage will be expanded to additional departments and user groups.

5 Approaches to Solving the Problem

5.1 Manual Centralization of Websites

A straightforward solution would involve manually integrating all existing university portals into a single, centralized website.

Advantages: Simple architectural concept and No need for complex AI algorithms.

Disadvantages: Requires extensive coordination among departments and High maintenance costs and limited scalability.

5.2 Keyword-Based Search Engine

Another possible approach is developing an internal search engine that indexes all university websites and retrieves pages based on keyword matches . This approach is currently being implemented at Ben Gurion, but it does not solve the problem well, and therefore this approach was once accepted and is currently undergoing adaptation.

Advantages: Easy to implement using existing web indexing tools and Provides quick answers to simple and infrequent queries (when there aren't many results for a word).

Disadvantages: Does not understand context or intent behind natural-language questions , Returns many irrelevant results and Produces irrelevant or incomplete results when terminology differs from official phrasing.

5.3 Direct Usage of General LLMs (Zero-Shot Generation)

A naive approach would be to rely on general-purpose LLMs (like ChatGPT or Claude) without any external architecture. Users would simply input their questions directly into the model interface.

Advantages: Zero development cost, immediate availability, and high conversational fluency.

Disadvantages: Dependency on Specific Context (The "URL Paradox"): General models cannot autonomously browse or crawl the entire university domain to find an answer. They require the user to provide a specific URL or copy-paste relevant text. This defeats the purpose of the system: it assumes the user already knows where the information is located. If

a student is lost and doesn't know which department handles their request, they cannot provide the correct source, rendering the tool useless for navigation.
Lack of Domain Knowledge: General models are trained on public internet data up to a specific cutoff. They do not have access to Ben-Gurion University's internal documents, specific PDFs, or recently updated regulations hidden in administrative sub-pages.
Hallucinations: Without access to the specific source text, the model will likely "invent" a logical-sounding but factually incorrect procedure (e.g., inventing a generic registration form deadline that doesn't exist at BGU).
Language Barriers: While capable in Hebrew, general models often struggle with the specific administrative terminology and acronyms used uniquely within Israeli academia.

5.4 AI-Driven Natural Language Interface (Selected Approach)

The chosen approach, Retrieval-Augmented Generation , combines Large Language Models (LLMs) with Vector Search to create an intelligent semantic search system. The system converts website content into embeddings representing the meaning of each segment. When a user submits a query, it is converted into an embedding and matched against stored vectors to find relevant content. Finally, an LLM formulates a concise, context-aware response based strictly on these retrieved chunks.
Advantages: Deep Understanding: Understands natural language and user intent, providing accurate results even without exact keyword matches.
Grounded Accuracy: Unlike general LLMs (discussed in Section 5.3), this system bases its answers strictly on retrieved university documents, significantly reducing hallucinations.
Source Transparency: Provides direct links to authoritative sources for user verification.
Scalability: Adaptable to additional university systems and data sources.
Disadvantages: Requires computing resources for embedding and model inference.
Needs periodic retraining and data synchronization to keep the index fresh.
Requires more development effort than a standard Keyword-Based Search Engine.

5.5 Off-the-Shelf Commercial Solutions

Several commercial tools currently offer AI-based search and chatbot functionalities that could theoretically assist in navigating large digital ecosystems.
Advantages: These commercial solutions are mature, reliable, and provide ready-to-use integration for large organizations with lots of money.
Disadvantages: They are costly, require vendor-managed infrastructure, and are not adaptable to academic environments such as universities. Specifically, they cannot easily integrate with systems like course registration, Moodle, or faculty websites at Ben-Gurion University.

6 Proposed Solution

The proposed solution focuses on the design and implementation of AskUni, an AI-driven intelligent navigation assistant engineered to streamline student interaction with the university's decentralized digital ecosystem. The project addresses the limitations of existing lexical (keyword-based) search engines by introducing a semantic search capability, allowing users to query administrative and academic databases using natural language.

At the core of the system is a Retrieval-Augmented Generation architecture. This architectural choice is critical for academic environments as it decouples the reasoning capabilities of Large Language Models (LLMs) from their training data. Instead of relying on static, pre-trained knowledge which may be outdated or prone to "hallucinations," the system generates answers by dynamically retrieving real-time context from a curated university knowledge base.

The solution is implemented via a two-phase pipeline. The first phase, Data Ingestion, involves an automated crawler that aggregates data from disparate sources, including pages and regulations. A specialized preprocessing module cleans this data and employs an algorithm to preserve the structure of tabular information (such as tuition rates or course schedules) before converting the text into high-dimensional vector embeddings stored in a Vector Database.

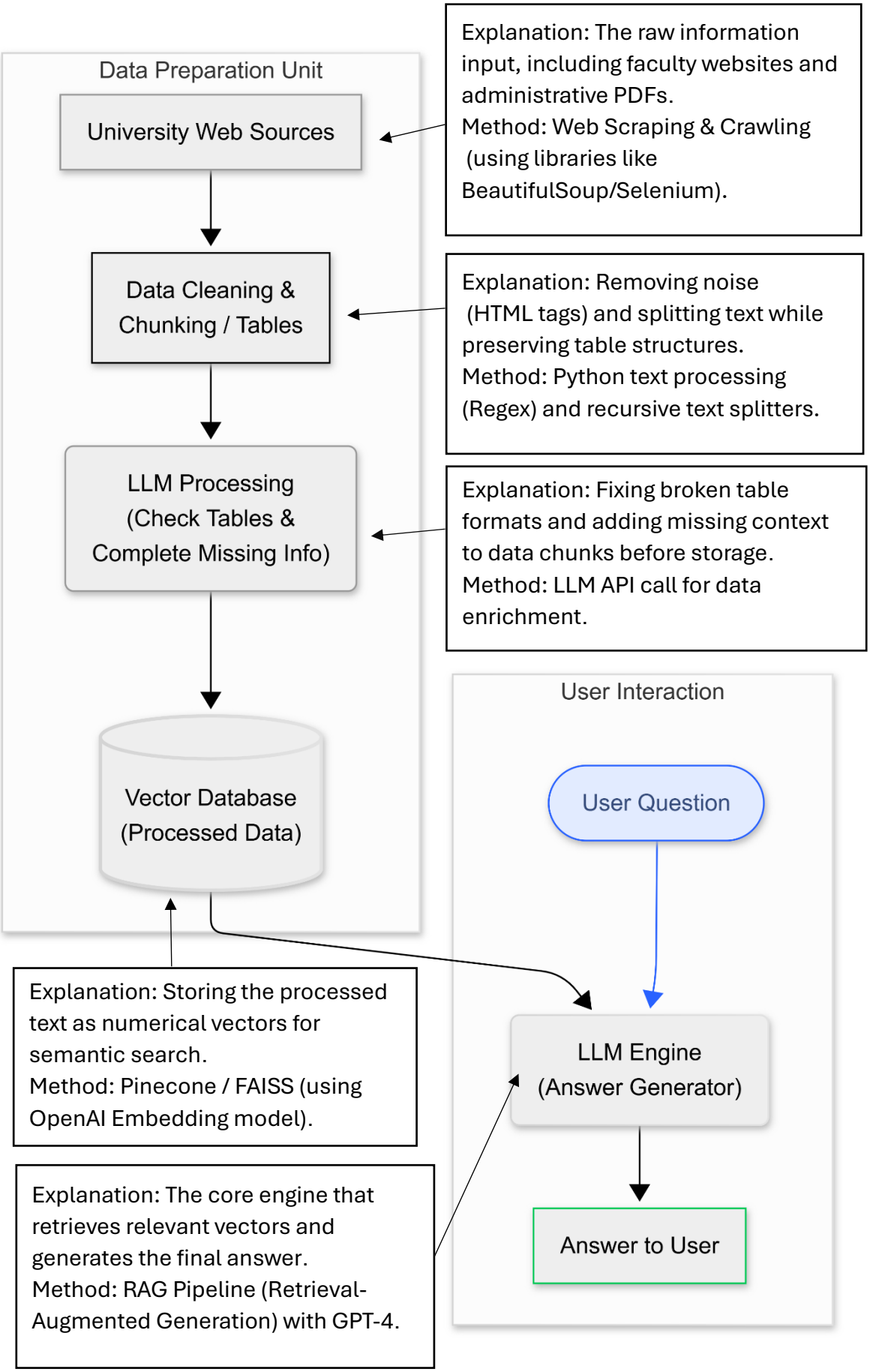
In the second phase, Runtime Interaction, the system processes user queries by converting them into vectors and performing a similarity search (using metrics like Cosine Similarity) to identify the most relevant document chunks. These chunks are injected as context into the LLM, which acts as a reasoning engine to synthesize a precise, fact-based response. This approach ensures that every answer is grounded in official university documents, providing students with a reliable and transparent source of information.

7 Design Proposal

This section outlines the detailed engineering design of the proposed solution. It begins with a technical breakdown of the system architecture and data flow, followed by an analysis of the project's operational constraints, working assumptions, and a preliminary risk assessment.

7.1 Technical Approach:

Architectural Diagram of the Proposed Solution and Step-by-Step Description:



7.2 Project Constraints The design process is subject to the following limitations:

Budgetary Constraints: The project operates on a student budget. Therefore, we utilize cost-effective APIs (e.g. OpenAI gpt-40-mini) and open-source libraries to minimize operational costs.

Data Privacy (GDPR/BGU Policy): The system is strictly prohibited from accessing or processing private student data (grades, passwords, medical records). The scope is limited to publicly available university information.

Compute Resources: The system relies on cloud-based inference (API) rather than local GPU training, due to the lack of high-end hardware availability.

Time Constraints: The functional prototype must be ready for the final presentation by the end of the second semester (June 2026).

7.3 Project Assumptions The proposed design relies on the following assumptions:

Site Structure Stability: We assume that the university's website architecture and internal HTML structure will remain consistent throughout the project's development. We assume no major overhauls or complete rebuilds of the website will occur, as such fundamental changes would require a significant redevelopment of the data ingestion pipeline.

Web Accessibility: It is assumed that Ben-Gurion University websites do not implement aggressive anti-bot blocking mechanisms (CAPTCHAs) that would permanently prevent data scraping for educational purposes.

API Stability: We assume that the selected LLM provider (e.g., OpenAI) maintains service availability and does not deprecate key models during the development lifecycle.

Language Support: We assume the underlying LLM has sufficient proficiency in Hebrew to understand academic administrative terminology without requiring fine-tuning.

7.4 Initial Risk Assessment

Risk: LLM Hallucinations (Accuracy)

Issue: The model may generate incorrect answers or "hallucinate" facts.

Mitigation: Enforce strict system prompts to rely only on retrieved context and return "I don't know" when information is missing.

Risk: High System Latency

Issue: Multiple API calls (Embedding → Retrieval → Generation) may cause slow response times.

Mitigation: Optimize chunk sizes, use asynchronous API calls, and select lightweight embedding models.

Risk: Semantic Loss in Embedding
Issue: The conversion of Hebrew academic text into vector embeddings may result in semantic loss, where the specific nuance of a query is missed by the embedding model.
Mitigation: Implement a Hybrid Search mechanism (combining vector search with keyword matching) or utilizing the LLM's large context window to process raw text directly for complex queries, bypassing the vector retrieval.

8 Project scope definition

The scope of AskUni is defined by the deliverables and the boundaries of what the project will and will not provide, and planning and evaluation apply only to what is explicitly included. in scope is a working prototype that accepts natural-language questions and returns grounded answers with supporting urls, an indexed knowledge base built from a selected set of official university web sources (public and/or explicitly permitted), and a repeatable retrieval pipeline that can be rerun to refresh the content (collection, cleaning, chunking, embedding, and indexing). also in scope are the response rules that enforce grounding and define how the system behaves when evidence is insufficient, an evaluation package consisting of a representative test set and reported results for metrics such as accuracy and grounding, retrieval quality, robustness, and latency, and documentation describing the architecture, the included data sources, the update strategy, and known limitations. the external interfaces include a simple user interface (cli or web) for entering questions and viewing answers and urls, the defined list of university domains or pages used as ingestion inputs, and the output format of answer text plus urls (and optionally short supporting snippets). explicitly out of scope are access to private student records or personal data, performing authenticated actions such as submitting forms or registering courses, guaranteeing answers for information not present in the indexed authoritative sources, and immediate full coverage of every university system or subdomain beyond the selected sources defined for the project. when a question depends on content outside the indexed sources, AskUni does not invent an answer; it returns the closest relevant authoritative links available and states that the retrieved sources do not fully support a definitive response.

9 Final Verification and Validation

Overview The goal of the final testing phase is to validate that the "AskUni" system meets the performance and functional requirements defined in the Specification Sheet. The testing process will evaluate the system's accuracy, response speed (latency), and stability under load.
Test Methodology To ensure objective results, the testing will be divided into three main

categories:

1) Accuracy Testing (The "Ground Truth" Test)

Dataset: We will compile a Benchmark Dataset consisting of 100 pairs of Questions and Correct Answers (Ground Truth). The dataset will cover diverse university topics: Course registration, administrative forms, contact details, and academic regulations.

Procedure: Each question from the dataset will be fed into the system. The output will be compared to the Ground Truth by a human evaluator and marked as "Correct" or "Incorrect".

Note: Questions regarding information not present in the ingested documents (out-of-domain) will be excluded from the accuracy calculation.

2) Performance & Stability Testing:

Latency Measurement: We will measure the round-trip time for each question in the benchmark set to calculate the average system latency. The goal is to ensure no query exceeds the maximum defined wait time.

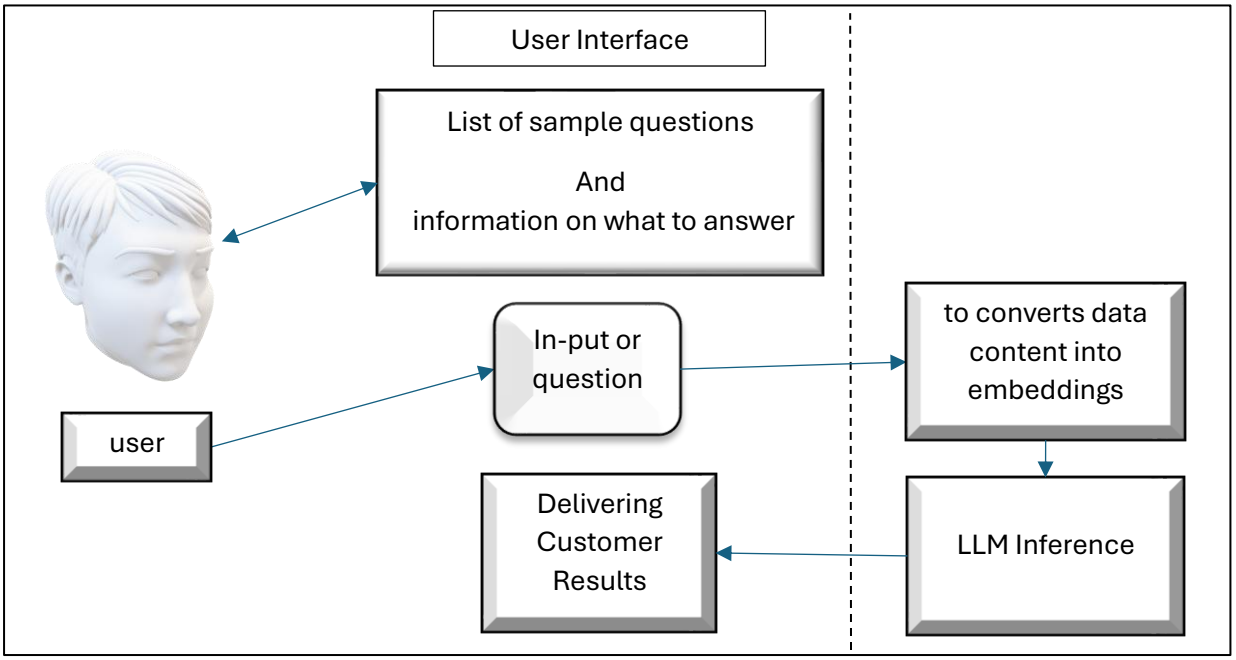
Load Testing: We will simulate concurrent access by multiple users (e.g., 5 simultaneous connections) to validate that the server remains stable, does not crash, and maintains acceptable response times under load.

3) User Satisfaction Surveys We will conduct user trials to evaluate the subjective experience. The surveys will focus on two key metrics:

Efficiency: Does the bot provide answers faster than manual searching.

Usability: Is the bot intuitive and easy to operate.

10 Flowchart illustration



11 Technology and innovations to solve the problem

The project is built on advanced AI technologies, including Large Language Models (LLMs) for natural language understanding and Vector Search for semantic information retrieval. Additionally, it leverages modern Python libraries for data collection, processing, and integration across the university's decentralized systems.

Together, these technologies enable intelligent, efficient, and context-aware navigation within the academic digital ecosystem.

Project Innovation: The core innovation is not just the use of AI, but its specific application:

Intent-Based Search: Transforming the current keyword-based search into a semantic, conversational system that understands user intent.

Platform Unification: Creating a single, intelligent interface that operates on top of the university's fragmented digital systems, saving users time.

A Custom & Cost-Effective Alternative: Unlike expensive, generic commercial products, this project introduces an open, cost-effective solution specifically optimized for BGU's academic content and workflows.

12 Budget Estimation

Development & Research (2 Students): 18,000$

Academic Supervisor : 6000$

Overhead: 25% of personnel costs ($24,000 × 0.25) = 6,000$

AI Infrastructure (Copilot/Tools): Cost: $60/month × 6 months = 360$

AI API Usage (Gemini 1.5 Pro): Cost: Estimated 100$ /month × 6 months = 600$

Cloud Hosting (Backend): Cost: $50/month × 6 months = 300$

Purchase of a .com or .co.il domain for one year = 20$

Large-format poster printing = 75$

UI/UX Assets: Purchase of a premium dashboard template or icon pack = 120$

AI Model Evaluation & QA = 200$

Vector Database Hosting: Utilization of a managed service ($40/month × 6 months) = 240$

User Testing Incentives = 200$

Search Accuracy Optimization = 65$

Legal & Compliance: Flat estimate for consultation = 320$

Miscellaneous: Contingencies (Unforeseen Expenses) = 1000$

Total Estimated Project Value = 33,500$

13 Literature review

Universities increasingly operate through large, distributed digital ecosystems that include public-facing departmental sites, course catalogs, registration systems, student portals, and administrative pages maintained by different units. While this expansion improves access to information, it also fragments it: students often do not know which system "owns" a process, which page is authoritative, or what exact terminology the institution uses for a given requirement. In this setting, traditional keyword-based search performs well only when users already know the official phrasing; it frequently fails when users ask in natural language, use synonyms, or need an answer that is implicitly spread across multiple pages. AskUni is motivated by this reality: it aims to provide a unified natural-language interface that can retrieve relevant institutional sources and answer questions with direct links to authoritative pages, reducing time spent searching and reducing the likelihood of following outdated or irrelevant guidance.

A common earlier approach to academic assistants is the domain-specific FAQ or intent-based chatbot, often implemented as an intent classifier paired with predefined responses. Sayyed et al. [1] exemplify this direction with a "Smart FAQ" chatbot focused on enrollment-related questions and enhanced with a voice-assistant interface. Their deployment emphasizes the operational attractiveness of guided dialogue: when the domain is narrow and the knowledge base is curated, the system can respond quickly and consistently, and the authors report strong measured performance in their setting [1]. However, this paradigm scales imperfectly in a university-wide context. The reliance on predefined intents and response templates makes coverage expansion expensive, especially when policies and procedures vary across departments and evolve over time. Moreover, the moment users ask questions that combine multiple intents ("where do I submit X and by when, and who approves it?") or use unexpected phrasing, purely intent-driven pipelines tend to either fail gracefully by redirecting users to generic pages or fail ungracefully by returning incorrect canned responses. These limitations point to the need for retrieval-based grounding and broader knowledge acquisition strategies if the target is cross-department, cross-platform university navigation rather than a single narrow workflow.

More recent deployments explore generative chatbots as a way to reduce the rigidity of fixed FAQs while supporting more natural interaction. Abdulla et al. [2] describe a generative AI chatbot integrated into an engineering scientific journal website, motivated by the desire to automate repetitive inquiries and reduce staff workload. Their analysis over a multi-month

period illustrates that such systems can provide meaningful operational value when they answer common questions and direct users to the right resources [2]. At the same time, their work highlights issues that become central as soon as generation is introduced: response reliability depends strongly on how the system is instructed, on what content it is allowed to use as ground truth, and on the existence of monitoring and refinement practices. In institutional contexts—where a wrong answer can lead to missed deadlines, incorrect submissions, or administrative mistakes—governance becomes a first-class requirement. The literature therefore supports a design stance in which the model is constrained to authoritative sources, is encouraged to admit uncertainty when evidence is insufficient, and is iteratively improved through feedback and content updates rather than treated as "set-and-forget" automation [2]. For AskUni, this is directly relevant: the assistant must behave as a navigation and guidance layer grounded in sources, not as an unconstrained conversational agent that might confidently improvise.

Another strand of work aims to address scale by automating knowledge acquisition and structuring website content for conversational access. Hsih et al. [3] propose an intelligent web service agent that leverages a website crawler to collect domain information and build a dialogue tree, combined with a neural NLU component (Seq2Seq with BiLSTM and attention) to improve intent and entity understanding. Their results suggest that learning-based NLU can provide robust improvements over simpler baselines and can outperform general NLU frameworks in certain in-domain settings [3]. This direction is valuable because it treats institutional content as something that can be systematically collected and organized rather than hand-authored into a knowledge base. However, crawler-plus-dialogue-tree systems tend to be site-specific: the dialogue structure and the agent's "world model" are shaped by a particular portal's navigation hierarchy and content conventions. Universities often lack a single uniform structure; instead, they consist of multiple independently managed sites with inconsistent navigation, mixed content types, and varying update frequency. In such environments, maintaining dialogue-tree representations per site can become a maintenance challenge similar to (or worse than) maintaining curated FAQ intents. This suggests that while automated crawling is an important ingestion mechanism, a less brittle representation than site-specific dialogue trees is desirable when the assistant must operate across heterogeneous sources.

This need for cross-source robustness motivates the recent emphasis on semantic retrieval and retrieval-augmented generation (RAG). Jing et al. [4] survey how large language models can be combined with vector databases to support similarity-based retrieval, external memory,

semantic caching, and grounded generation. The central promise of this architecture is that users can express needs in natural language while retrieval finds relevant content even when the user's wording differs from institutional phrasing; the system no longer depends on an exhaustive list of intents or hand-maintained responses. For AskUni's target use case, these properties matter: a student might ask "how do I appeal a grade," "who approves an exception," or "where do I submit a special request," and the assistant must locate relevant passages across multiple pages that may not contain those exact keywords. Embedding-based retrieval offers a principled mechanism for bridging vocabulary mismatches, while the vector index provides a scalable substrate for searching large and evolving corpora [4]. At the same time, the survey implicitly underscores that such systems are not "free": chunking strategy, metadata design, index updates, reranking, latency, and evaluation methodology all materially influence whether the assistant produces reliable outcomes or simply produces fluent answers with weak evidence. Therefore, an AskUni-like system must treat retrieval and grounding as core engineering components and not as an afterthought added to generation.

Beyond direct question answering, LLM-based academic assistants also target recommendation and guidance tasks where a user's need is exploratory rather than purely factual. Van Deventer et al. [5] present an LLM approach to course recommendations that uses natural-language queries and an embedding-based retrieval pipeline over course descriptions, including a technique in which the system generates an "ideal course description" from the user's interests and retrieves real courses similar to it in vector space [5]. While course recommendation is a narrower domain than university-wide navigation, it demonstrates a broader principle: LLMs can translate informal user intent into a representation that supports effective retrieval, and the retrieval layer can then ground suggestions in a concrete institutional corpus. This is relevant to AskUni because many student questions are mixed-mode: they involve both factual constraints ("deadline," "eligibility," "required documents") and decision support ("which office," "which procedure," "what should I do first"). A retrieval-augmented assistant can, in principle, support both—provided that it remains constrained by authoritative sources and clearly signals where it is summarizing versus where it is recommending.

Taken together, these sources suggest a progression in the design space. Intent-based FAQ bots can achieve high performance in narrow, curated domains but scale poorly across fragmented institutional ecosystems and diverse phrasing [1]. Generative chatbots can improve conversational flexibility and reduce manual response authoring, but they introduce governance challenges and require explicit grounding and continuous oversight to maintain

trustworthiness [2]. Crawler-driven dialogue agents can automate content acquisition and improve NLU robustness, yet they tend to remain tied to site-specific structures that are difficult to maintain across heterogeneous university sites [3]. The emerging LLM-plus-vector-database paradigm offers a scalable foundation for cross-source semantic retrieval and grounded generation, aligning naturally with the needs of a multi-platform university environment [4]. Finally, LLM-driven retrieval pipelines can support richer academic assistance scenarios such as recommendations when grounded in institutional descriptions and constraints [5].

AskUni is positioned at the intersection of these insights: it aims to unify fragmented university information access through a retrieval-augmented architecture that indexes authoritative sources semantically and produces answers grounded in retrieved evidence, with direct links that allow users to verify and act. The literature points to several requirements that shape this positioning. First, the assistant should minimize dependence on fixed intent taxonomies and instead use retrieval to adapt to evolving content and diverse user phrasing, addressing the scaling limitations observed in FAQ-style deployments [1]. Second, it must incorporate governance practices—grounding, uncertainty handling, and iterative improvement—to avoid the reliability pitfalls of unconstrained generation noted in generative deployments [2]. Third, it should benefit from automated ingestion (crawling/collection) without committing to brittle per-site dialogue trees, acknowledging the maintenance challenges of site-structured conversational agents in heterogeneous environments [3]. Fourth, it should follow established best practices in LLM–vector database integration, treating chunking, metadata, reranking, refresh, and evaluation as core components that determine real-world quality [4]. Finally, it should remain extensible toward more guided support experiences—such as recommending the “next best page” or “relevant procedure”—while ensuring that suggestions remain traceable to institutional sources, as illustrated by recommendation-focused academic assistants [5].

In summary, existing work provides strong evidence that conversational interfaces can improve access to institutional information, but it also shows that success depends on grounding, maintainability, and scalability across evolving content. AskUni addresses a gap in this landscape by targeting cross-platform university navigation rather than a single domain or single portal, and by adopting a retrieval-augmented design that aims to combine the flexibility of natural-language interaction with the reliability of source-based evidence and direct links.

14 Bibliography

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המלצת ציון לדו"ח מכין

<u>אם יש צורך, לכל סטודנט/ית בנפרד</u>

מספר הפרויקט: P-2026-037 , שם הפרויקט:

שם המנחה : דן וילאנציק

שם הסטודנט/ית: דורון ויצמן גיל פרדמן ת.ז:

%	קריטריון	1 – חלש	2 – בינוני	3 – טוב	4 – ט"ב	5 - מצוין
12	הגדרת המטרה - האם מטרת הפרויקט ברורה? המטרה צריכה לכלול גם את התרומה הצפויה מהשלמת הפרויקט.					
13	הבנת הבעיה - האם ניכר שהסטודנט הבין את הבעיה ואת הפתרונות הקיימים? האם נערך סקר ספרות רלוונטי? (חיפוש מקורות והבנת עבודות דומות)					
12	הארכיטקטורה המוצעת - האם הגישה הכללית ורכיבי הפתרון הוצגו והוסברו בצורה ברורה? האם ניכרת הבנה? האם הובהר מדוע הארכיטקטורה המוצעת לפתרון ההנדסי מתאימה לפתרון ההנדסי?					
13	הצגת הדו"ח המכין - האם יש מבנה הגיוני לדו"ח המכין ? האם איכות ההצגה טובה? האם ישנה התייחסות לכל חלקי הדו"ח ? האם יש קשר הגיוני בין החלקים? האם רמת התחבר טובה ?					
12	רמת קושי - כיצד הנך מעריך את רמת הקושי הצפויה בפרויקט בהשוואה לפרויקטים אחרים השנה או בשנים קודמות?					
13	מדדים לבדיקת הצלחת הפרויקט - האם הוגדרו קריטריונים ברורים ומדידים כמותית לבחינת הצלחת הפרויקט? האם הקריטריונים שהוצגו תואמים להגדרת מטרות הפרויקט?					
12	איכות הפרויקט המוצע - עבור פרויקטים מחקריים- מהי רמת הצעת מחקר? עבור פרויקטים טכני ים - מהי רמת המפרט טכני? האם הדרישות עונות על הצורך?					
13	המאפיינים העיקריים בפרויקט - יש לסמן את כל המאפיינים הרלוונטיים שהושגו (ולו באופן חלקי). מאפיין אחד בלבד יזכה במחצית הניקוד. שני מאפיינים ומעלה יזכו בניקוד מלא.					

הערכת רמת הקושי של הפרויקט: קל מאוד / קל / בינוני / קשה / קשה מאוד

הערות:_____